

Data Cleaning and Preparation using Power BI Power Query Editor

Name : Prisha Jaiswal

AIML B

23070126095

Objective

The goal of this task was to clean and prepare a dataset for visualization in Power BI by applying essential preprocessing steps using Power Query Editor. The following operations were performed:

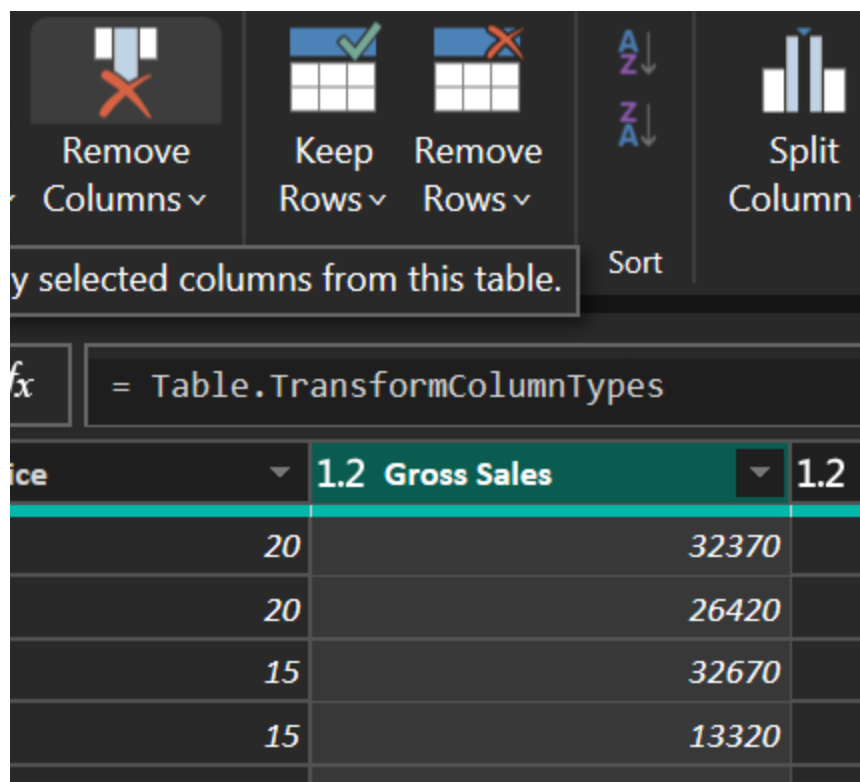
1. Remove Unnecessary Columns
2. Handle Nulls and Missing Data
3. Rename Columns
4. Change Data Types
5. Remove Duplicates
6. Create Calculated Columns (extract month, year, etc.)
7. Merge or Append Queries, Sort and Filter
8. Save and Apply Changes
9. Document Applied Steps

1. Remove Unnecessary Columns

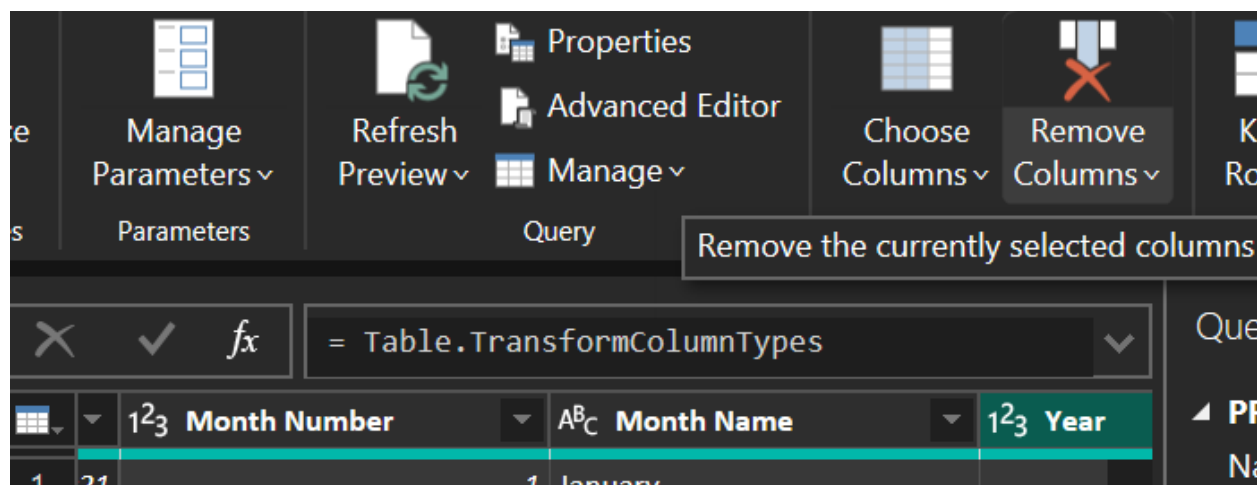
To streamline the dataset, I removed the following columns:

- Gross Sales: This can be derived from existing columns using $\text{Units Sold} \times \text{Sale Price}$, making it redundant.
- Month Number, Month Name, Year: These can be extracted from the Date column using built-in functions in Power BI.

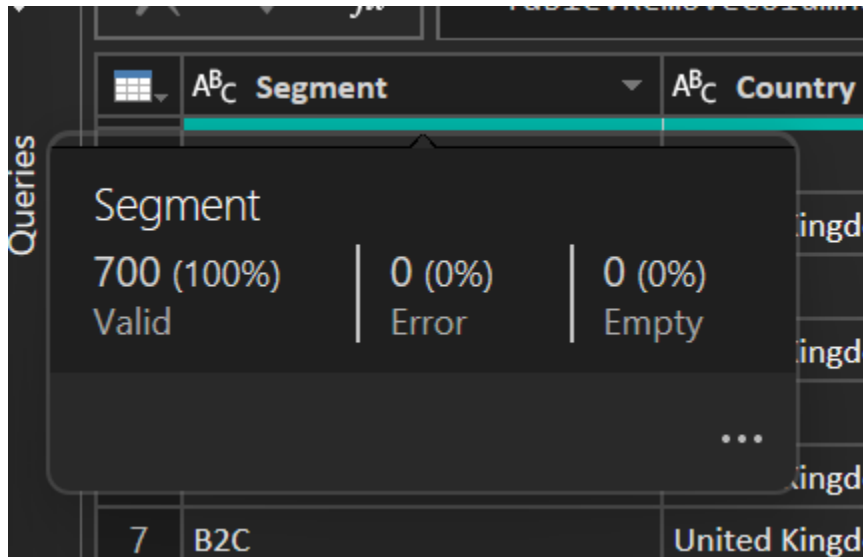
Removing these columns reduces clutter and simplifies downstream analysis.



Month Number and Month Name and Year (we can extract these from the Date column if needed)



2. Handle Nulls and Missing Data



The screenshot shows a Power Query interface with a table containing columns 'Segment' and 'Country'. A tooltip for the 'Segment' column is displayed, showing the following data:

Segment		
700 (100%)	0 (0%)	0 (0%)
Valid	Error	Empty

The background table shows the following data:

	Segment	Country
7	B2C	United Kingdom

Null or empty values can lead to inaccurate analysis and errors in visuals. There are two main ways to detect them:

- Hover over the data preview in Power Query to spot nulls.
- Use the dropdown filter in each column to check if (null) appears.

Based on the context of the column, I used different approaches:

- Remove Empty Rows: This option was used when entire rows were empty.
- Filter or Replace Nulls: For specific columns, missing values were either filtered out or replaced appropriately.

The Remove Empty feature under the Home → Remove Rows section was used to clean fully empty records.

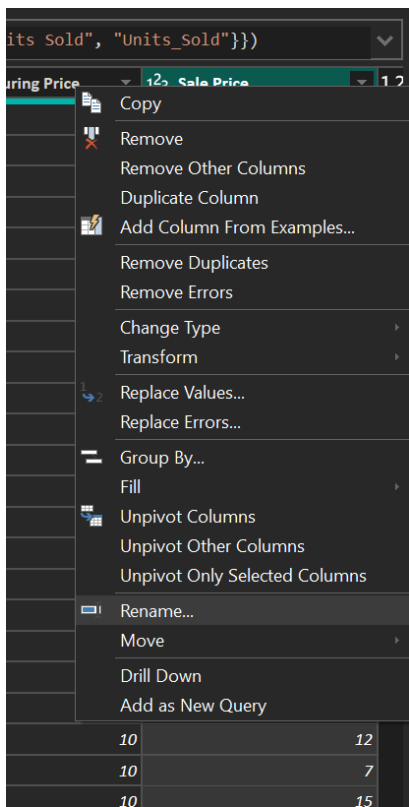
The remove empty will help remove the null/empty columns

3. Rename Columns

Rename columns to be clean and readable. Column names were updated to improve readability and follow a consistent naming convention. Some examples include:

- Units Sold → Units_Sold
- Sale Price → Sale_Price
- Manufacturing Price → Mfg_Price
- COGS → Cost Of Goods Sold

These changes help simplify formulas and enhance clarity when creating visuals or measures.



Preview						
Parameters						
Columns						
Advanced						
Dependencies						
fx = Table.RenameColumns(#"Removed Columns",{{"Units Sold", "Units_Sold"}, {"Sale Price", "Sale_Price"}, {"Manufacturing Price", "Mfg_Price"}}						
	Product	Units_Sold	Mfg_Price	Sale_Price	Gross Sales	
	Doohickey	1618.5	3	20	32370	
	Doohickey	1221	2	20	26420	

	1.2 Cost of Goods Sold	1.2 Profit
32370	16185	
26420	1221	

4. Change Data Types

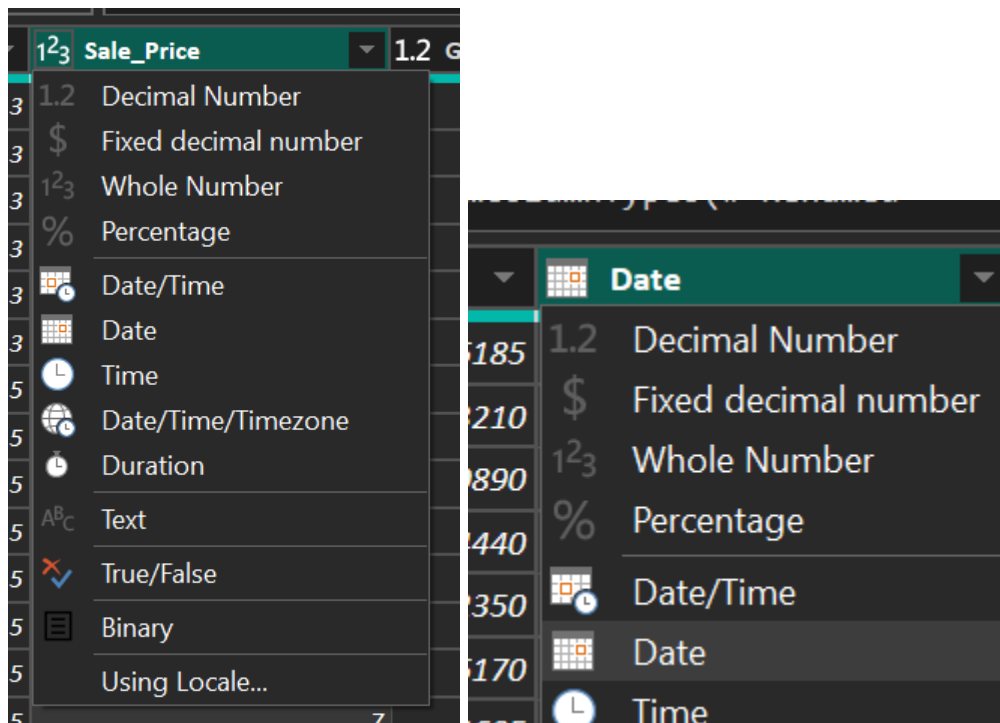
To ensure accurate calculations and sorting, I reviewed and corrected data types:

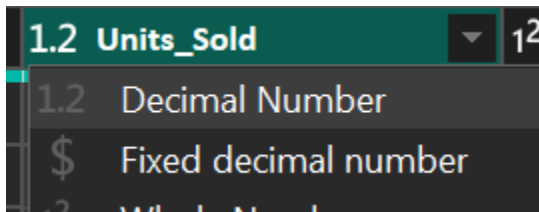
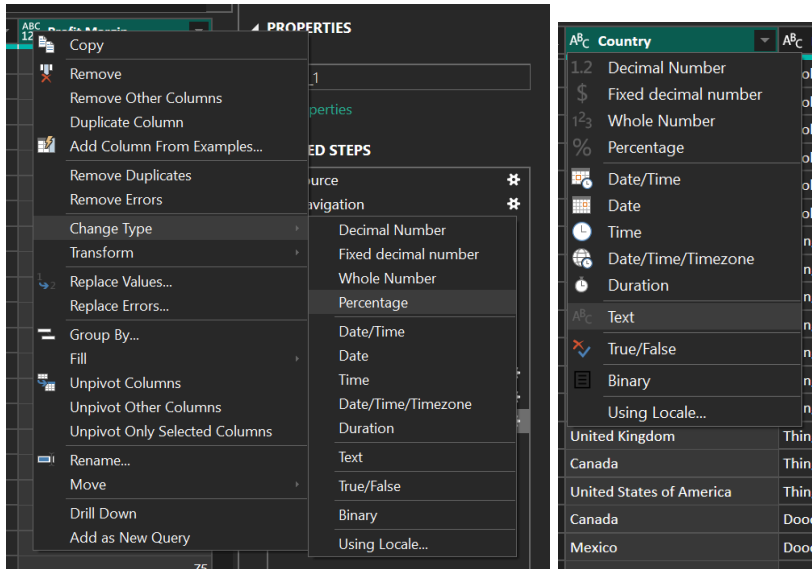
- Date → Date
- Units_Sold, COGS, Profit, Sale_Price → Decimal Number
- Segment, Country, Product → Text

Changing data types avoids issues with sorting, aggregations, and visual consistency.

How:

1. Click on the small data type icon next to the column name.
2. Choose the correct type.

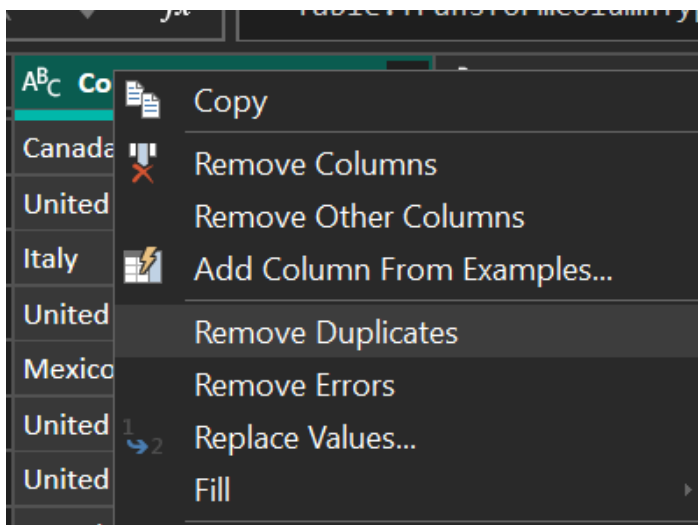




5: Remove Duplicates

How:

1. Select all columns(CTRL+A → Right-click → Remove Duplicates.



When working with datasets in Power BI, it's easy to get confused by repeated entries — especially if you notice countries or other categories appearing more than once. In my case, I saw multiple instances of the same country in the data. Naturally, I wondered: *Should I remove duplicates?*

The Reality: Not All Duplicates Are Bad

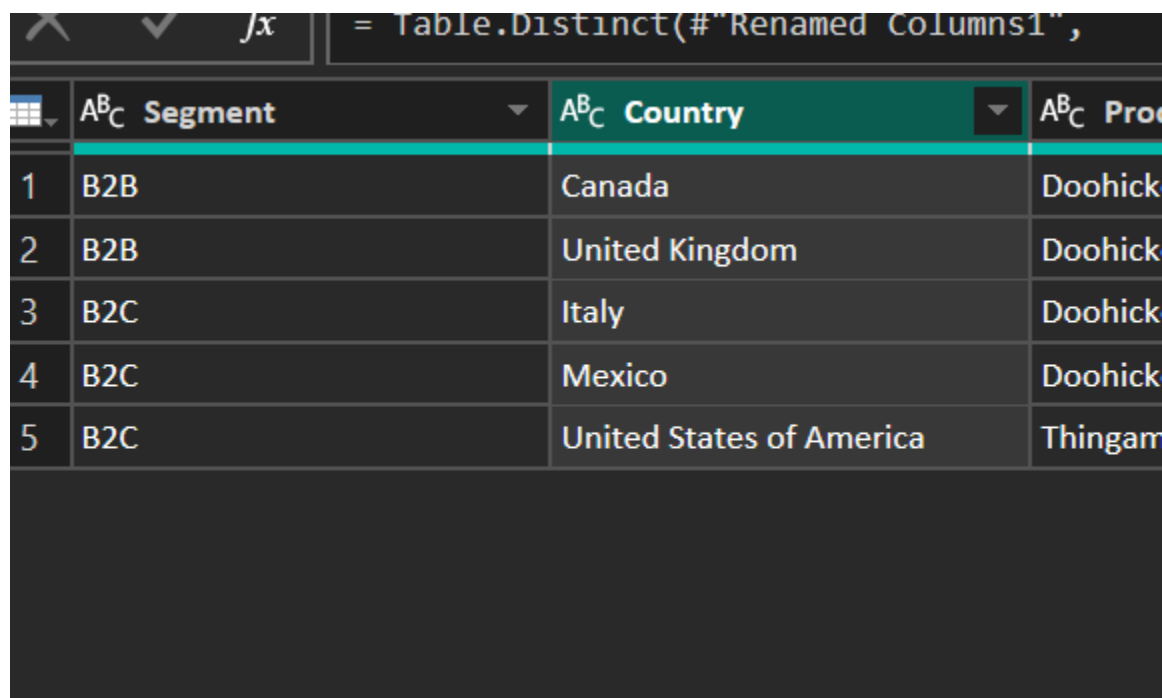
After closer inspection, I realized that these "duplicate" countries had different:

- Dates
- Products
- Segments
- Sales or Profit values

These rows weren't true duplicates — they were individual sales transactions. Each one represented a unique business record, even if some values like Country or Product repeated.

Result?

Only about 10 rows remained! That's because Power BI kept only the rows where *every single column was unique*. This clearly deleted a majority of valid and important rows.



The screenshot shows a Power BI interface with a formula bar at the top containing the DAX formula: `= Table.Distinct(#"Renamed Columns1",`. Below the formula bar is a table with three columns: **Segment**, **Country**, and **Product**. The table contains five rows of data, each with a unique combination of these three attributes.

	Segment	Country	Product
1	B2B	Canada	Doohick
2	B2B	United Kingdom	Doohick
3	B2C	Italy	Doohick
4	B2C	Mexico	Doohick
5	B2C	United States of America	Thingam

6: Create Calculated Columns

Date-Based Fields

To enhance temporal analysis, I created new columns:

- Year: Extracted from the Date column using Add Column → Date → Year
- Month Name: Extracted from the Date column using Add Column → Date → Month → Name of the Month

These fields allow for better sorting and time-based aggregations.

Profit Margin Calculation

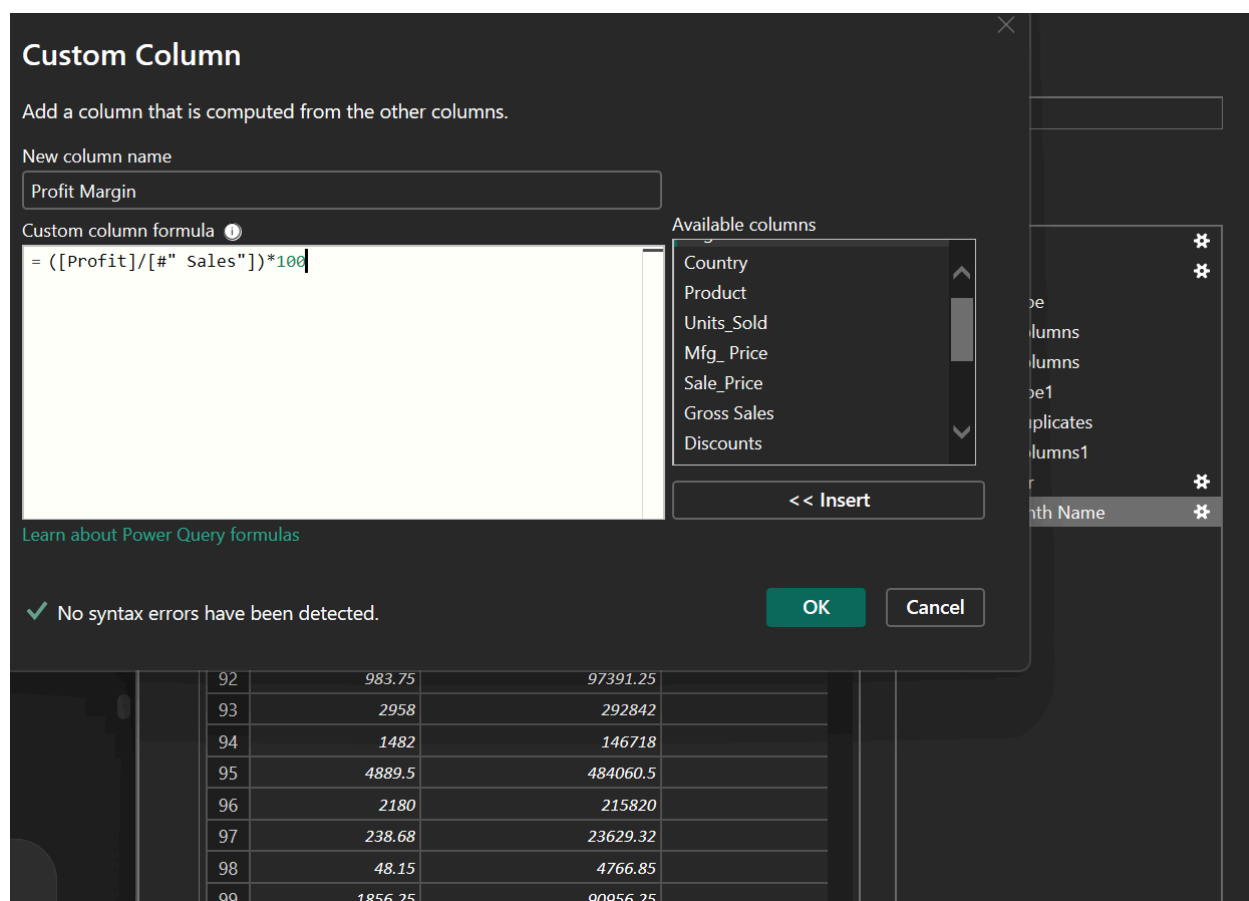
I introduced a new calculated column:

$$\text{Profit Margin (\%)} = (\text{Profit} / \text{Sales}) * 100$$

In Power Query, this was created via:

The screenshot shows the Power BI Desktop interface. On the left, the 'Date' field list is visible, showing a list of dates from 01-01-2021 to 01-09-2020. On the right, the 'Date' field dropdown menu is open, showing a list of date-related fields: Year, Start of Year, End of Year, Month, Start of Month, End of Month, Days in Month, Name of Month, Quarter of Year, Start of Quarter, End of Quarter, Week of Year, Week of Month, Start of Week, End of Week, Day, Day of Week, Day of Year, and Start of Day. The 'Name of Month' field is highlighted.

Date	1 ² ₃ Year	A ^B _C Month Name
01-01-2021	2021	January
01-01-2021	2021	January
01-06-2021	2021	June



This metric is useful to evaluate profitability across different products, countries, or segments.

7. Merge or Append Queries, Sort and Filter

In this task, the dataset consisted of a single table, so no merge or append operations were necessary. All transformations were done within the same query.

However, I applied sorting on the Date column in ascending order to maintain chronological consistency for future trend analysis.

Filters were also applied where needed, such as removing rows with nulls or filtering out specific values in numeric columns.

= Table.Sort(#"Inserted Month Name",{{"Date", Order.Ascending}})							
Price	1.2 Discounts	1.2 Sales	1.2 COGS	1.2 Profit	Date	APC	Month Name
350	12600	239400	187200	52200	01-09-2020		September
350	0	534450	397020	137430	01-09-2020		September
7	369.6	11950.4	8800	3150.4	01-09-2020		September

Help



Refresh

Preview

▼



Properties



Advanced Editor



Manage

▼

Query



Choose Columns

▼



Remove Columns

▼

Manage Columns



Keep Rows

▼



Remove Rows

▼

Reduce Rows



Sort

Sort lowest to highest.

▼



Split Column

▼



Group By



Use First Row



Replace Value

▼

Transform

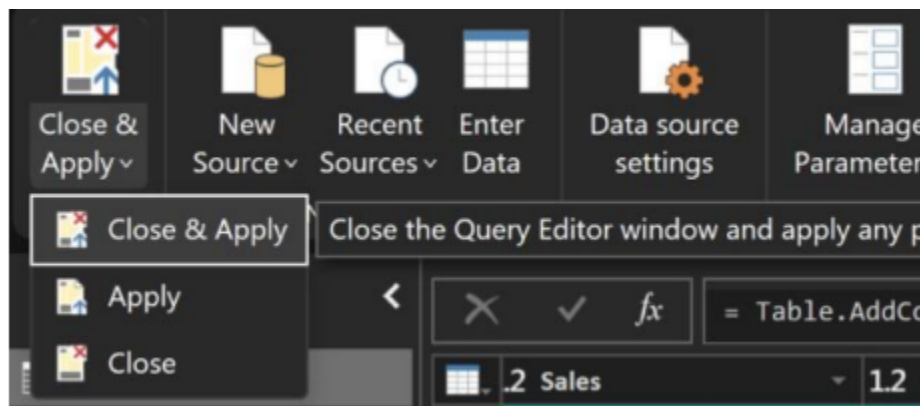
Data Type:

▼

```
ate", Order.Ascending}})
```

▼	1.2 Profit	 Date	 1 ² 3 Year	AB _C Month Name	▼	1.2
7200	52200	01-09-2020	2020	September		
7020	137430	01-09-2020	2020	September		
8800	3150.4	01-09-2020	2020	September		
1745	3875.85	01-09-2020	2020	September		
2045	3468.96	01-09-2020	2020	September		
3640	-8286.25	01-09-2020	2020	September		
4500	117406	01-09-2020	2020	September		

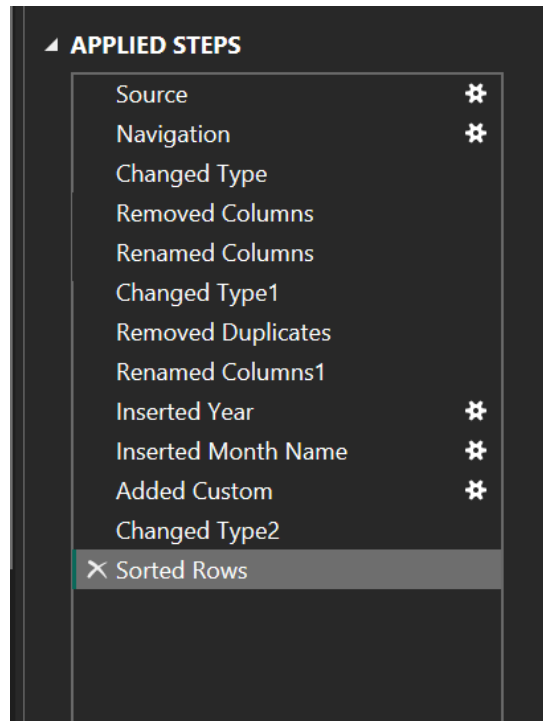
8. Saving your Progress In order to save your changes, in 'Home' tab, select 'Close & Apply'.



9. Snapshot of "Applied Steps" pane from Query Editor

Throughout the cleaning process, every action taken was automatically recorded in the Applied Steps pane of Power Query Editor. This pane serves as a step-by-step history of all transformations applied to the data.

Screenshots were taken after each major step to document the changes and can be provided along with this report.



Conclusion

The dataset is now clean, structured, and ready for visualization in Power BI. By removing redundancies, handling missing data, standardizing column names and types, and generating new analytical fields, the data has been transformed into a powerful asset for deriving business insights.